

Coordinate Geometry

CHAPTER 3

Pair of Straight Lines

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3

Pair of Straight Lines

EQUATION OF PAIR OF STRAIGHT LINES

If equations of two lines are $L_1 \equiv a_1x + b_1y + c_1 = 0$ and $L_2 \equiv a_2x + b_2y + c_2 = 0$, then their joint equation is expressed as $(a_1x + b_1y + c_1)(a_2x + b_2y + c_2) = 0$. This is the equation of pair of straight lines having component straight lines L_1 and L_2 .

For example, the joint equation of lines $2x + y + 3 = 0$ and $x - y + 4 = 0$ is $(2x + y + 3)(x - y + 4) = 0$ or $2x^2 - xy - y^2 + 11x + y - 12 = 0$.

Conversely, if combined equation of two straight lines or equation of pair of straight lines is $6x^2 + 5xy - 4y^2 = 0$, then this equation can be reduced to $(2x - y)(3x + 4y) = 0$. So, component lines are $2x - y = 0$ and $3x + 4y = 0$.

Note:

In order to find the combined equation of two lines, make the RHS of each equation of straight line equal to zero and then multiply the two equations.

ILLUSTRATION 3.1

Find the equations of two straight lines whose combined equation is $6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0$.

Sol. The given equation of pair of straight lines is

$$6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0$$

$$\text{or } 6x^2 + (5y + 7)x - (4y^2 - 13y + 3) = 0$$

Solving it as a quadratic in x , we get

$$\begin{aligned} x &= \frac{-(5y + 7) \pm \sqrt{(5y + 7)^2 + 24(4y^2 - 13y + 3)}}{12} \\ &= \frac{-(5y + 7) \pm \sqrt{121y^2 - 242y + 121}}{12} \\ &= \frac{-(5y + 7) \pm 11(y - 1)}{12} \\ &= \frac{6y - 18}{12}, \frac{-16y + 4}{12} \text{ or } \frac{y - 3}{2}, \frac{-4y + 1}{3} \end{aligned}$$

The two straight lines are $2x - y + 3 = 0$ and $3x + 4y - 1 = 0$.

ILLUSTRATION 3.2

Find the distance between the pair of parallel lines $x^2 + 4xy + 4y^2 + 3x + 6y - 4 = 0$.

Sol. Given lines are

$$(x + 2y)^2 + 3(x + 2y) - 4 = 0$$

$$\therefore x + 2y = \frac{-3 \pm \sqrt{9 + 16}}{2} = \frac{-3 \pm 5}{2} = -4, 1$$

Therefore, the lines are

$$x + 2y + 4 = 0$$

$$x + 2y - 1 = 0$$

$$\begin{aligned} \text{Required distance} &= \frac{|4 - (-1)|}{\sqrt{1 + 4}} \\ &= \frac{5}{\sqrt{5}} = \sqrt{5} \end{aligned}$$

GENERAL EQUATION OF THE SECOND DEGREE IN TWO VARIABLES

The most general form of a quadratic equation in variables x and y is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (1)$$

Since it is an equation in variables x and y , it must represent the equation of a locus in a plane. It may represent a pair of straight lines, circle, parabola, ellipse or hyperbola. If it represents pair of straight lines, then expression on L.H.S. must get factorized into two linear expressions. Let us find the condition for this.

If $a \neq 0$, we can write equation (1) as quadratic in x .

$$ax^2 + 2x(hy + g) + (by^2 + 2fy + c) = 0$$

Solving for x , we get

$$\begin{aligned} x &= \frac{-2(hy + g) \pm \sqrt{4(hy + g)^2 - 4a(by^2 + 2fy + c)}}{2a} \\ &= \frac{-(hy + g) \pm \sqrt{(h^2 - ab)y^2 + 2(gh - af)y + (g^2 - ac)}}{a} \end{aligned}$$

We must have x as linear expression in y . For that, the expression under the square root must be a perfect square of some linear expression in y . Thus, the discriminant of expression $(h^2 - ab)y^2 + 2(gh - af)y + (g^2 - ac)$ must be zero.

$$\text{So, } 4(gh - af)^2 - 4(h^2 - ab)(g^2 - ac) = 0$$

$$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad (2)$$

This is condition for which equation (1) represents the pair of straight lines.

Equation (2) can be put in the form of determinant as follows:

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

Point of Intersection of Pair of Straight Lines

The elementary method of finding the point of intersection of pair of straight lines is getting the component straight lines and solving them.

We can also use the direct formula for the point of intersection which is $\left(\frac{bg - hf}{h^2 - ab}, \frac{af - gh}{h^2 - ab} \right)$.

The point of intersection can also be determined with the help of partial differentiation as follows:

Differentiating (1) w.r.t. x , keeping y constant, we get

$$2ax + 2hy + 2g = 0 \quad (3)$$

Now, differentiating (1) w.r.t. y , keeping x constant, we get

$$2hx + 2by + 2f = 0 \quad (4)$$

Now, solving equations (3) and (4), we get the point of intersection.

It should be noted that equations (3) and (4) are not equations of component straight lines.

ILLUSTRATION 3.3

Find the value of λ if $2x^2 + 7xy + 3y^2 + 8x + 14y + \lambda = 0$ represents a pair of straight lines.

Sol. The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of lines if $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.

So, for given equation,

$$6\lambda + 2(7)(4)\left(\frac{7}{2}\right) - 2(7)^2 - 3(4)^2 - \lambda\left(\frac{7}{2}\right)^2 = 0$$

$$\text{or } 6\lambda + 196 - 98 - 48 - \frac{49\lambda}{4} = 0$$

$$\text{or } \frac{49\lambda}{4} - 6\lambda = 196 - 146 = 50$$

$$\text{or } \frac{25\lambda}{4} = 50$$

$$\text{or } \lambda = \frac{200}{25} = 8$$

ILLUSTRATION 3.4

Does equation $x^2 + 2y^2 - 2\sqrt{3}x - 4y + 5 = 0$ satisfies the condition $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$? Does it represent a pair of straight lines?

Sol. Given equation is $x^2 + 2y^2 - 2\sqrt{3}x - 4y + 5 = 0$.

Here, $a = 1, b = 2, h = 0, g = -\sqrt{3}, f = -2, c = 5$.

But, $abc + 2fgh - af^2 - bg^2 - ch^2 = 10 + 0 - 4 - 6 = 0$.

The given equation can be written as

$$(x - \sqrt{3})^2 + 2(y - 1)^2 = 0$$

Hence, it denotes only a point $P(\sqrt{3}, 1)$ but not a pair of straight lines.

ILLUSTRATION 3.5

If the pair of lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ intersect on the y -axis, then prove that $2fgh = bg^2 + ch^2$.

Sol. Let the line intersect on y -axis at $P(0, y_1)$.

Putting this point in the equation of pair of straight lines, we get

$$by_1^2 + 2fy_1 + c = 0$$

Above equation must have equal roots.

$$\therefore 4f^2 - 4bc = 0$$

$$\text{or } f^2 = bc \quad (1)$$

Also, given equation represents pair of straight lines.

$$\therefore abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad (2)$$

From (1), putting the value of f^2 in (2), we get

$$2fgh = bg^2 + ch^2$$

ILLUSTRATION 3.6

Find the coordinates of points where pair of lines given by equation $2x^2 - 6y^2 + xy - 2x + 17y - 12 = 0$ intersect line $x = 1$.

Sol. Let the given pair of straight lines intersect line $x = 1$ at $(1, y)$.

Putting this point in the given equation, we get

$$2 - 6y^2 + y - 2 + 17y - 12 = 0$$

$$\Rightarrow y^2 - 3y + 2 = 0$$

$$\Rightarrow (y - 1)(y - 2) = 0$$

$$\Rightarrow y = 1, y = 2$$

Thus, required points of intersection are $(1, 1)$ and $(1, 2)$.

ILLUSTRATION 3.7

If the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents the pair of parallel straight lines, then prove that $h^2 = ab$ and $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.

Sol. Let the given equation represent pair of parallel straight lines $lx + my + n = 0$ and $lx + my + n' = 0$

$$\therefore ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

$$= (lx + my + n)(lx + my + n')$$

$$= (lx + my)^2 + l(n + n')x + m(n + n')y + nn' = 0$$

Thus, expression $(lx + my)^2$ is same as $ax^2 + by^2 + 2hxy$.

Therefore, $ax^2 + 2hxy + by^2$ must be perfect square of liner expression in x and y .

$$\text{So, } (2h)^2 - 4ab = 0$$

$$\therefore h^2 = ab$$

$$\text{Also, } abc + 2hgf - af^2 - bg^2 - ch^2 = 0.$$

CONCEPT APPLICATION EXERCISE 3.1

- Find the combined equation of the pair of lines through the point $(1, 0)$ and parallel to the lines represented by $2x^2 - xy - y^2 = 0$.
- Prove that the equation $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$ represents a pair of straight lines. Find the coordinates of their point of intersection and also the angle between them.
- If one of the lines of the pair $ax^2 + 2hxy + by^2 = 0$ bisects the angle between the positive direction of the axes, then find the relation for a, b , and h .
- If the pair of lines $\sqrt{3}x^2 - 4xy + \sqrt{3}y^2 = 0$ is rotated about the origin by $\pi/6$ in the anticlockwise sense, then find the equation of the pair of lines in the new position.

5. If the equation $2x^2 + kxy + 2y^2 = 0$ represents a pair of real and distinct lines, then find the values of k .
6. Find the point of intersection of the pair of straight lines represented by the equation $6x^2 + 5xy - 21y^2 + 13x + 38y - 5 = 0$.

ANSWERS

1. $2x^2 - xy - y^2 - 4x + y + 2 = 0$ 2. $(1, -2), \tan^{-1} 1/5$
 3. $a + b = -2h$ 4. $\sqrt{3}x^2 - xy = 0$
 5. $k \in (-\infty, -4) \cup (4, \infty)$ 6. $(-32/23, 17/32)$

SECOND-DEGREE HOMOGENEOUS EQUATION

The homogenous equation is that equation in which each term has same degree. General second-degree homogenous equation is given by

$$ax^2 + by^2 + 2hxy = 0, h^2 > ab \quad (1)$$

This equation represents pair of straight lines passing through origin.

Consider one such equation of pair of straight lines.

$$(2x - y)(3x + 2y) = 0$$

$$\text{or } 6x^2 - 2y^2 + xy = 0 \quad (2)$$

Now, consider another equation of pair of straight lines.

$$(2x - y + 3)(3x + 2y - 4) = 0$$

$$\text{or } 6x^2 - 2y^2 + xy + x + 10y - 12 = 0 \quad (3)$$

The component lines of above equation are parallel to the component lines of equation (2).

We observe that second degree terms in both the equations (2) and (3) are same or ratio of coefficients of second-degree terms is same.

Now, general equation of pair of straight line is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (4)$$

Then corresponding equation of pair of straight lines parallel to component lines of above equation and passing through origin is given by

$$ax^2 + by^2 + 2hxy = 0 \quad (5)$$

Let the component lines of equation (5) be $y - m_1x = 0$ and $y - m_2x = 0$.

$$\begin{aligned} \therefore by^2 + 2hxy + ax^2 &= b(y - m_1x)(y - m_2x) \\ &= by^2 - b(m_1 + m_2)xy + bm_1m_2x^2 \end{aligned}$$

Comparing coefficients of both sides, we get

$$2h = -b(m_1 + m_2) \text{ and } a = bm_1m_2$$

$$\therefore m_1 + m_2 = \frac{-2h}{b} \text{ and } m_1m_2 = \frac{a}{b}$$

ILLUSTRATION 3.8

Find the equations of component lines whose combined equation is $6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0$ without solving for x or y .

Sol. Given equation of pair of straight lines is:

$$6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0 \quad (1)$$

Homogenous part of given equation is

$$6x^2 + 5xy - 4y^2 = 0 \quad (2)$$

$$\text{or } (3x + 4y)(2x - y) = 0$$

This represents equations of pair of straight lines parallel to the component lines of given equation.

Therefore, given equation of pair of straight lines can be put as

$$(3x + y + A)(2x - y + B) = 0 \quad (3)$$

Comparing coefficients of (1) and (3), we get

$$A = -1 \text{ and } B = 3$$

Hence, the required component lines are $2x - y + 3 = 0$, $3x + 4y - 1 = 0$.

ILLUSTRATION 3.9

If the component lines whose combined equation is $px^2 - qxy - y^2 = 0$ make the angles α and β with x -axis, then find the value of $\tan(\alpha + \beta)$.

Sol. Given equation of pair of straight lines is:

$$px^2 - qxy - y^2 = 0 \quad (1)$$

Let the component lines be $y = m_1x$ and $y = m_2x$.

$$\therefore m_1 = \tan \alpha \text{ and } m_2 = \tan \beta \quad (\text{Given})$$

$$\text{Also, } m_1 + m_2 = -q \text{ and } m_1m_2 = -p$$

$$\begin{aligned} \therefore \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{m_1 + m_2}{1 - m_1m_2} = \frac{-q}{1 + p} \end{aligned}$$

ILLUSTRATION 3.10

Find the joint equation of pair of lines which passes through origin and are perpendicular to the lines represented by the equation $y^2 + 3xy - 6x + 5y - 14 = 0$.

Sol. Homogeneous part of the given equation is $y^2 + 3xy = 0$. So, component lines are $y = 0$ and $y + 3x = 0$.

Lines perpendicular to these lines are $x = 0$ and $x - 3y = 0$.

So, combined equation of above lines is $x^2 - 3xy = 0$.

ILLUSTRATION 3.11

If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then find the value of c .

Sol. Given equation of pair of lines is $x^2 - 2cxy - 7y^2 = 0$.

Let the slopes of the component lines be m_1 and m_2 .

It is given that

$$\begin{aligned} m_1 + m_2 &= 4m_1m_2 \\ \Rightarrow -\frac{2c}{7} &= \frac{-4}{7} \\ \therefore c &= 2 \end{aligned}$$

ILLUSTRATION 3.12

If the pair of straight lines $ax^2 + 2hxy + by^2 = 0$ is rotated about the origin through 90° , then find its equation in the new position.

Sol. Given equation of pair of straight lines is

$$ax^2 + 2hxy + by^2 = 0.$$

Let the component lines be $y = m_1x$ and $y = m_2x$.

$$\therefore m_1 + m_2 = \frac{-2h}{b} \text{ and } m_1m_2 = \frac{a}{b}$$

Now, equations of lines perpendicular to above component lines are

$$y = -\frac{1}{m_1}x \text{ and } y = -\frac{1}{m_2}x$$

$$\text{or } m_1y + x = 0 \text{ and } m_2y + x = 0$$

Combined equation of above lines is

$$(m_1y + x)(m_2y + x) = 0$$

$$\text{or } m_1m_2y^2 + xy(m_1 + m_2) + x^2 = 0$$

$$\text{or } \frac{a}{b}y^2 + xy\left(\frac{-2h}{b}\right) + x^2 = 0$$

$$\text{or } bx^2 - 2hxy + ay^2 = 0$$

Alternative method:

We can write the given equation of pair of straight lines as:

$$b\left(\frac{y}{x}\right)^2 + 2h\left(\frac{y}{x}\right) + a = 0$$

The roots of this equation are m_1 and m_2 which are slopes of component lines.

Now, we require equation whose roots are $-\frac{1}{m_1}$ and $-\frac{1}{m_2}$.

So, in above equation, replacing $\frac{y}{x}$ by $-\frac{x}{y}$, we get

$$b\left(-\frac{x}{y}\right)^2 + 2h\left(-\frac{x}{y}\right) + a = 0$$

$$\text{or } bx^2 - 2hxy + ay^2 = 0$$

which is the required equation of pair of straight lines.

ANGLE BETWEEN PAIR OF STRAIGHT LINES

Consider general equation of pair of straight lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (1)$$

The corresponding equation of pair of straight lines parallel to component lines of above equation and passing through origin is

$$ax^2 + 2hxy + by^2 = 0 \quad (2)$$

Clearly, angle between component lines of (1) and (2) is same.

Let the component lines of equation (2) be $y - m_1x = 0$ and $y - m_2x = 0$.

$$\therefore m_1 + m_2 = \frac{-2h}{b} \text{ and } m_1m_2 = \frac{a}{b}$$

If angle between lines is θ , then

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1m_2} \right| = \frac{\sqrt{(m_1 + m_2)^2 - 4m_1m_2}}{|1 + m_1m_2|} \\ &= \frac{\sqrt{\frac{4h^2}{b^2} - 4\frac{a}{b}}}{\left|1 + \frac{a}{b}\right|} = \frac{2\sqrt{h^2 - ab}}{|a + b|} \end{aligned}$$

This is the formula for finding the acute angle between component lines of equation (1) or (2).

Special cases:

If lines are parallel or coincident, then $\theta = 0$. So, $h^2 = ab$.

If lines are perpendicular, then $\theta = 90^\circ$. So, $a + b = 0$.

ILLUSTRATION 3.13

Find acute and obtuse angle between component lines whose combined equation is $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$.

Sol. For the given equation of pair of straight lines, we have $a = 2$, $b = 3$, $h = 5/2$

Thus, acute angle between lines is given by

$$\theta = \tan^{-1} \frac{2\sqrt{h^2 - ab}}{|a + b|} = \frac{2\sqrt{\frac{25}{4} - 6}}{2 + 3} = \tan^{-1} \frac{1}{5}$$

$$\text{Obtuse angle between lines} = \left(\pi - \tan^{-1} \frac{1}{5} \right)$$

ILLUSTRATION 3.14

Find the value of a for which the lines represented by $ax^2 + 5xy + 2y^2 = 0$ are mutually perpendicular.

Sol. The lines given by $ax^2 + 5xy + 2y^2 = 0$ are mutually perpendicular if $a + 2 = 0$, i.e., $a = -2$.

EQUATION OF PAIR OF ANGLE BISECTORS OF GIVEN PAIR OF STRAIGHT LINES

Let the given equation of pair of straight lines is

$$ax^2 + by^2 + 2hxy = 0 \quad (1)$$

Let the component lines be $y = m_1x$ and $y = m_2x$.

$$\therefore m_1 + m_2 = \frac{-2h}{b}, m_1m_2 = \frac{a}{b}$$

Equations of angle bisectors of component lines are

$$\frac{y - m_1x}{\sqrt{1 + m_1^2}} = \pm \frac{y - m_2x}{\sqrt{1 + m_2^2}}$$

$\therefore$ Combined equation of bisector lines is

$$\left(\frac{y - m_1x}{\sqrt{1 + m_1^2}} - \frac{y - m_2x}{\sqrt{1 + m_2^2}} \right) \left(\frac{y - m_1x}{\sqrt{1 + m_1^2}} + \frac{y - m_2x}{\sqrt{1 + m_2^2}} \right) = 0$$

$$\text{or } \frac{(y - m_1x)^2}{1 + m_1^2} - \frac{(y - m_2x)^2}{1 + m_2^2} = 0$$

$$\text{or } (m_1 + m_2)(x^2 - y^2) + 2(m_1m_2 - 1)xy = 0$$

$$\text{or } \frac{-2h}{b}(x^2 - y^2) + 2\left(\frac{a}{b} - 1\right)xy = 0$$

$$\text{or } \frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

ILLUSTRATION 3.15

If the pair of straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ are such that each pair bisects the angle between the other pair, then prove that $pq = -1$.

Sol. Combined equation of the bisectors of the angles between pair of lines $x^2 - 2pxy - y^2 = 0$ is

$$\frac{x^2 - y^2}{1 - (-1)} = \frac{xy}{-p}$$

or $px^2 + 2xy - py^2 = 0$

This must be same as the given equation of pair of bisectors i.e., $x^2 - 2qxy - y^2 = 0$.

Comparing ratios of coefficients, we get

$$\frac{p}{1} = \frac{1}{-q} = \frac{p}{1}$$

$\therefore pq = -1$

HOMOGENIZATION OF SECOND DEGREE EQUATION

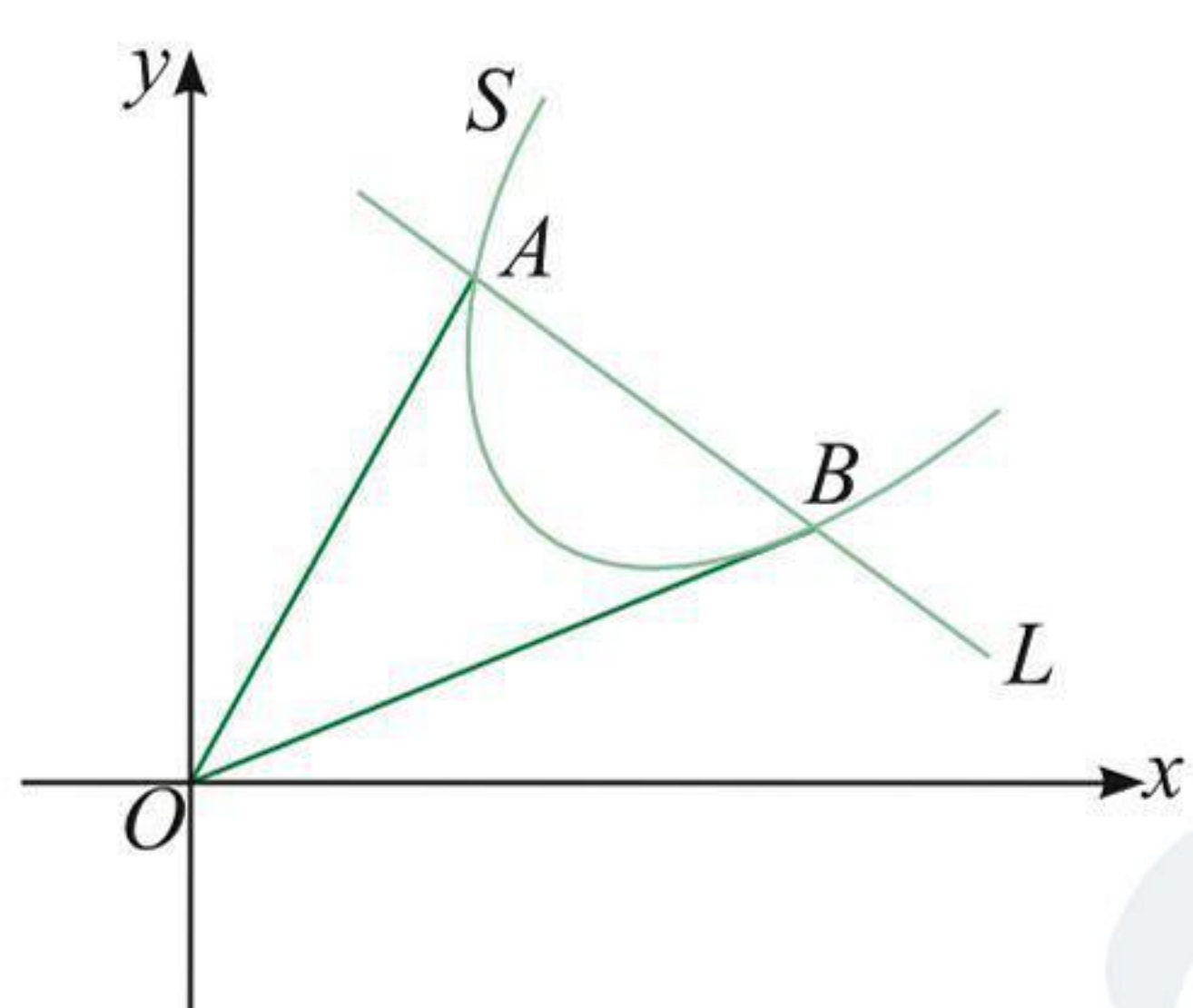
Consider general second degree equation

$$S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (1)$$

This equation represents pair of straight lines, circle, parabola, ellipse or hyperbola.

Clearly, this is not homogenous equation.

Now, let the line $L \equiv lx + my + n = 0$ intersect the curve (1) in two points A and B .



The combined equation of pair of straight lines OA and OB must be homogenous and must contain only second-degree terms.

From the equation of line, we have

$$1 = \frac{lx + my}{-n} \quad \dots(2)$$

Now, equation of curve is

$$(ax^2 + 2hxy + by^2) + (2gx + 2fy) \cdot 1 + c \cdot 1^2 = 0$$

Putting the value of '1' from (2) in above equation, we have

$$ax^2 + 2hxy + by^2 + (2gx + 2fy) \left(\frac{lx + my}{-n} \right) + c \left(\frac{lx + my}{-n} \right)^2 = 0$$

This is homogenous equation of second degree.

Since points A and B satisfy above equation, it represents the pair of straight lines OA and OB . This process of making a second-degree equation homogeneous is called homogenization through which we get equation of pair of straight lines joining the points of intersection of the given line and curve with the origin.

ILLUSTRATION 3.16

Prove that the straight lines joining the origin to the points of intersection of the straight line $hx + ky = 2hk$ and the curve $(x - k)^2 + (y - h)^2 = c^2$ are at right angle if $h^2 + k^2 = c^2$.

Sol. From the equation of line, we have

$$1 = \frac{hx + ky}{2hk} \quad (1)$$

Equation of the curve is

$$x^2 + y^2 - 2kx - 2hy + h^2 + k^2 - c^2 = 0$$

Making above equation homogeneous with the help of (1), we get

$$x^2 + y^2 - 2(kx + hy) \left(\frac{hx + ky}{2hk} \right) + (h^2 + k^2 - c^2) \left(\frac{hx + ky}{2hk} \right)^2 = 0$$

This is combined equation of the pair of lines joining the origin to the points of intersection of the given line and the curve. The component lines are perpendicular if sum of coefficient of x^2 and coefficient of y^2 is zero.

$$\therefore (h^2 + k^2)(h^2 + k^2 - c^2) = 0$$

$$\Rightarrow h^2 + k^2 = c^2 \quad (\text{as } h^2 + k^2 \neq 0)$$

ILLUSTRATION 3.17

Find the angle between the lines joining the origin to the points of intersection of the straight line $y = 3x + 2$ with the curve $x^2 + 2xy + 3y^2 + 4x + 8y - 11 = 0$.

Sol. The equation of straight line is

$$\frac{y - 3x}{2} = 1$$

Making given equation of curve homogeneous, we get

$$x^2 + 2xy + 3y^2 + (4x + 8y) \left(\frac{y - 3x}{2} \right) - 11 \left(\frac{y - 3x}{2} \right)^2 = 0$$

$$\text{or } 7x^2 - 2xy - y^2 = 0 \quad (1)$$

This is the equation of the pair of lines joining origin to the point of intersection of the given line and curve.

Comparing this equation with the equation $ax^2 + 2hxy + by^2 = 0$, we get

$$a = 7, b = -1 \text{ and } h = -1$$

If θ is the acute angle between component lines of (1), then

$$\tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right| = \left| \frac{2\sqrt{1 + 7}}{7 - 1} \right| = \frac{2\sqrt{8}}{6} = \frac{2\sqrt{2}}{3}$$

$$\therefore \theta = \tan^{-1} \frac{2\sqrt{2}}{3}$$

CONCEPT APPLICATION EXERCISE 3.2

1. If the slope of one line is double the slope of another line and the combined equation of the pair of lines is $(x^2/a) + (2xy/h) + (y^2/b) = 0$, then find the ratio $ab : h^2$.
2. Find the angle between the lines represented by $x^2 + 2xy \sec \theta + y^2 = 0$.

3. Find the angle between the straight lines joining the origin to the points of intersection of $3x^2 + 5xy - 3y^2 + 2x + 3y = 0$ and $3x - 2y = 1$.
4. If θ is the angle between the lines given by the equation $6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0$, then find the equation of the line passing through the point of intersection of these lines and making an angle θ with the positive x -axis.
5. Show that the equation of the pair of lines bisecting the angles between the pair of bisectors of the angles between the pair of lines $ax^2 + 2hxy + by^2 = 0$ is $(a - b)(x^2 - y^2) + 4hxy = 0$.

6. Find the equation of the bisectors of the angles between the lines joining the origin to the point of intersection of the straight line $x - y = 2$ with the curve $5x^2 + 11xy - 8y^2 + 8x - 4y + 12 = 0$.
7. Show that the pairs of straight lines $2x^2 + 6xy + y^2 = 0$ and $4x^2 + 18xy + y^2 = 0$ have the same set of angular bisector.

ANSWERS

- | | | |
|------------------------|---------------------------|------------|
| 1. $9 : 8$ | 2. θ | 3. $\pi/2$ |
| 4. $11x - 2y + 13 = 0$ | 6. $x^2 + 30xy - y^2 = 0$ | |

Exercises

Single Correct Answer Type

- The angle between the pair of lines whose equation is $4x^2 + 10xy + my^2 + 5x + 10y = 0$ is
 - $\tan^{-1}(3/8)$
 - $\tan^{-1}(3/4)$
 - $\tan^{-1}\{2\sqrt{25-4m}/(m+4)\}, m \in R$
 - none of these
- The two lines represented by $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ are perpendicular to each other for
 - two values of a
 - a
 - for one value of a
 - for no value of a
- The distance between the two lines represented by the equation $9x^2 - 24xy + 16y^2 - 12x + 16y - 12 = 0$ is
 - $8/5$
 - $6/5$
 - $11/5$
 - none of these
- The equations $x - y = 4$ and $x^2 + 4xy + y^2 = 0$ represent the sides of
 - an equilateral triangle
 - a right-angled triangle
 - an isosceles triangle
 - none of these
- The straight lines represented by $(y - mx)^2 = a^2(1 + m^2)$ and $(y - nx)^2 = a^2(1 + n^2)$ form a
 - rectangle
 - rhombus
 - trapezium
 - none of these
- If the pairs of lines $x^2 + 2xy + ay^2 = 0$ and $ax^2 + 2xy + y^2 = 0$ have exactly one line in common, then the joint equation of the other two lines is given by
 - $3x^2 + 8xy - 3y^2 = 0$
 - $3x^2 + 10xy + 3y^2 = 0$
 - $y^2 + 2xy - 3x^2 = 0$
 - $x^2 + 2xy - 3y^2 = 0$
- The condition that one of the straight lines given by the equation $ax^2 + 2hxy + by^2 = 0$ may coincide with one of those given by the equation $a'x^2 + 2h'xy + b'y^2 = 0$ is
 - $(ab' - a'b)^2 = 4(ha' - h'a)(bh' - b'h)$
 - $(ab' - a'b)^2 = (ha' - h'a)(bh' - b'h)$
 - $(ha' - h'a)^2 = 4(ab' - a'b)(bh' - b'h)$
 - $(bh' - b'h)^2 = 4(ab' - a'b)(ha' - h'a)$
- If the lines represented by the equation $3y^2 - x^2 + 2\sqrt{3}x - 3 = 0$ are rotated about the point $(\sqrt{3}, 0)$ through an angle of 15° , one in clockwise direction and the other in anticlockwise direction, so that they become perpendicular, then the equation of the pair of lines in the new position is
 - $y^2 - x^2 + 2\sqrt{3}x + 3 = 0$
 - $y^2 - x^2 + 2\sqrt{3}x - 3 = 0$
 - $y^2 - x^2 - 2\sqrt{3}x + 3 = 0$
 - $y^2 - x^2 + 3 = 0$
- The equation of a line which is parallel to the line common to the pair of lines given by $6x^2 - xy - 12y^2 = 0$ and $15x^2 + 14xy - 8y^2 = 0$ and at a distance of 7 units from it is
 - $3x - 4y = -35$
 - $5x - 2y = 7$
 - $3x + 4y = 35$
 - $2x - 3y = 7$
- The equation $x^2y^2 - 9y^2 + 6x^2y + 54y = 0$ represents
 - a pair of straight lines and a circle
 - a pair of straight lines and a parabola
 - a set of four straight lines forming a square
 - none of these
- The equations $a^2x^2 + 2h(a+b)xy + b^2y^2 = 0$ and $ax^2 + 2hxy + by^2 = 0$ represent
 - two pairs of perpendicular straight lines
 - two pairs of parallel straight lines
 - two pairs of straight lines which are equally inclined to each other
 - none of these
- If the equation of the pair of straight lines passing through the point $(1, 1)$, one making an angle θ with the positive direction of the x -axis and the other making the same angle with the positive direction of the y -axis, is $x^2 - (a+2)xy + y^2 + a(x+y-1) = 0, a \neq -2$, then the value of $\sin 2\theta$ is
 - $a - 2$
 - $a + 2$
 - $2/(a+2)$
 - $2/a$
- If two lines represented by $x^4 + x^3y + cx^2y^2 - xy^3 + y^4 = 0$ bisect the angle between the other two, then the value of c is
 - 0
 - 1
 - 1
 - 6
- Through a point A on the x -axis, a straight line is drawn parallel to the y -axis so as to meet the pair of straight lines $ax^2 + 2hxy + by^2 = 0$ at B and C . If $AB = BC$, then
 - $h^2 = 4ab$
 - $8h^2 = 9ab$
 - $9h^2 = 8ab$
 - $4h^2 = ab$
- The image of the pair of lines represented by $ax^2 + 2hxy + by^2 = 0$ by the line mirror $y = 0$ is
 - $ax^2 - 2hxy - by^2 = 0$
 - $bx^2 - 2hxy + ay^2 = 0$
 - $bx^2 + 2hxy + ay^2 = 0$
 - $ax^2 - 2hxy + by^2 = 0$
- The straight lines represented by the equation $135x^2 - 136xy + 33y^2 = 0$ are equally inclined to the line
 - $x - 2y = 7$
 - $x + 2y = 7$
 - $x - 2y = 4$
 - $3x + 2y = 4$
- If the slope of one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ is the square of the other, then $\frac{a+b}{h} + \frac{8h^2}{ab} =$
 - 4
 - 6
 - 8
 - none of these
- $x + y = 7$ and $ax^2 + 2hxy + ay^2 = 0, (a \neq 0)$, are three real distinct lines forming a triangle. Then the triangle is
 - isosceles
 - scalene
 - equilateral
 - right-angled

19. Statement 1: If $-2h = a + b$, then one line of the pair of lines $ax^2 + 2hxy + by^2 = 0$ bisects the angle between the coordinate axes in the positive quadrant.

Statement 2: If $ax + y(2h + a) = 0$ is a factor of $ax^2 + 2hxy + by^2 = 0$, then $b + 2h + a = 0$.

- (1) Both the statements are true but statement 2 is the correct explanation of statement 1.
 - (2) Both the statements are true but statement 2 is not the correct explanation of statement 1.
 - (3) Statement 1 is true and statement 2 is false.
 - (4) Statement 1 is false and statement 2 is true.
- 20.** The orthocenter of the triangle formed by the lines $xy = 0$ and $x + y = 1$ is
- (1) $(1/2, 1/2)$
 - (2) $(1/3, 1/3)$
 - (3) $(0, 0)$
 - (4) $(1/4, 1/4)$
- 21.** Let PQR be a right-angled isosceles triangle, right angled at $P(2, 1)$. If the equation of the line QR is $2x + y = 3$, then the equation representing the pair of lines PQ and PR is
- (1) $3x^2 - 3y^2 + 8xy + 20x + 10y + 25 = 0$
 - (2) $3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$
 - (3) $3x^2 - 3y^2 + 8xy + 10x + 15y + 20 = 0$
 - (4) $3x^2 - 3y^2 - 8xy - 15y - 20 = 0$
- 22.** Area of the triangle formed by the line $x + y = 3$ and the angle bisectors of the pairs of straight lines $x^2 - y^2 + 2y = 1$ is
- (1) 2 sq. units
 - (2) 4 sq. units
 - (3) 6 sq. units
 - (4) 8 sq. unit
- 23.** The orthocentre of the triangle formed by the lines $2x^2 + 3xy - 2y^2 - 9x + 7y - 5 = 0$ with $4x + 5y - 3 = 0$ is
- (1) $(3/5, 11/5)$
 - (2) $(6/5, 11/5)$
 - (3) $(5/6, 11/5)$
 - (4) $(3/5, 6/5)$

Multiple Correct Answers Type

- 1.** The equation $x^3 + x^2y - xy^2 = y^3$ represents
 - (1) three real straight lines
 - (2) lines in which two of them are perpendicular to each other
 - (3) lines in which two of them are coincident
 - (4) none of these
- 2.** The straight lines represented by $x^2 + mxy - 2y^2 + 3y - 1 = 0$ meet at
 - (1) $(-1/3, 2/3)$
 - (2) $(-1/3, -2/3)$
 - (3) $(1/3, 2/3)$
 - (4) none of these
- 3.** If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is
 - (1) 1
 - (2) 2
 - (3) $-1/2$
 - (4) -1
- 4.** If $x^2 + 2hxy + y^2 = 0$ ($h > 1$) represents the equation of the straight lines through the origin which make an angle α with the straight line $y + x = 0$, then
 - (1) $\sec 2\alpha = h$
 - (2) $\cos \alpha = \sqrt{(1+h)/(2h)}$
 - (3) $2 \sin \alpha = \sqrt{(1+h)/h}$
 - (4) $\cot \alpha = \sqrt{(h+1)/(h-1)}$

- 5.** The combined equation of three sides of a triangle is $(x^2 - y^2)(2x + 3y - 6) = 0$. If $(-2, a)$ is an interior point and $(b, 1)$ is an exterior point of the triangle, then
 - (1) $2 < a < 10/3$
 - (2) $-2 < a < 10/3$
 - (3) $-1 < b < 9/2$
 - (4) $-1 < b < 1$
- 6.** If one of the lines given by the equation $2x^2 + pxy + 3y^2 = 0$ coincide with one of those given by $2x^2 + qxy - 3y^2 = 0$ and the other lines represented by them are perpendicular, then value of $p + q$ is
 - (1) 6
 - (2) -6
 - (3) -7
 - (4) 7
- 7.** The lines joining the origin to the point of intersection of $3x^2 + mxy - 4x + 1 = 0$ and $2x + y - 1 = 0$ are at right angles. Then which of the following is a possible value of m ?
 - (1) -4
 - (2) 4
 - (3) 7
 - (4) 3
- 8.** If the equation $ax^2 - 6xy + y^2 + bx + cy + d = 0$ represents a pair of lines whose slopes are m and m^2 , then the value(s) of a is/are
 - (1) $a = -8$
 - (2) $a = 8$
 - (3) $a = 27$
 - (4) $a = -27$
- 9.** Two pairs of straight lines have the equations $y^2 + xy - 12x^2 = 0$ and $ax^2 + 2hxy + by^2 = 0$. One line will be common among them if
 - (1) $a + 8h - 16b = 0$
 - (2) $a - 8h + 16b = 0$
 - (3) $a - 6h + 9b = 0$
 - (4) $a + 6h + 9b = 0$

Linked Comprehension Type

For Problems 1–3

Consider the equation of a pair of straight lines as $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$.

- 1.** The value of λ is
 - (1) 1
 - (2) 2
 - (3) $3/2$
 - (4) 3
- 2.** The point of intersection of lines is (α, β) . Then the value of $\alpha\beta$ is
 - (1) 35
 - (2) 45
 - (3) 20
 - (4) 15
- 3.** The angle between the lines is θ . Then the value of $\tan \theta$ is
 - (1) $1/5$
 - (2) $2/9$
 - (3) $1/7$
 - (4) $3/4$

For Problems 4–6

Consider a pair of perpendicular straight lines $ax^2 + 3xy - 2y^2 - 5x + 5y + c = 0$.

- 4.** The value of a is
 - (1) 1
 - (2) 3
 - (3) 2
 - (4) -2
- 5.** The value of c is
 - (1) -3
 - (2) 3
 - (3) -1
 - (4) 1
- 6.** Distance between the orthocenter and the circumcenter of triangle ABC is
 - (1) 4
 - (2) $9/2$
 - (3) $8/3$
 - (4) $7/4$

Numerical Value Type

- The distance between the lines $(x + 7y)^2 + 4\sqrt{2}(x + 7y) - 42 = 0$ is _____.
- Area of the triangle formed by the lines $y^2 - 9xy + 18x^2 = 0$ and $y = 6$ is _____.
- The value k for which $4x^2 + 8xy + ky^2 = 9$ is the equation of a pair of straight lines is _____.

- If the gradient of one of the lines $x^2 + hxy + 2y^2 = 0$ is twice that of the other, then sum of possible values of h is _____.
- One of the bisectors of the angle between the lines $a(x - 1)^2 + 2h(x - 1)(y - 2) + b(y - 2)^2 = 0$ is $x + 2y - 5 = 0$. Then the value of $\frac{b - a}{h}$ is _____.

Answers Key

EXERCISES

Single Correct Answer Type

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (2) | 2. (1) | 3. (1) | 4. (1) | 5. (2) |
| 6. (2) | 7. (1) | 8. (2) | 9. (3) | 10. (3) |
| 11. (3) | 12. (3) | 13. (4) | 14. (2) | 15. (4) |
| 16. (2) | 17. (2) | 18. (1) | 19. (2) | 20. (3) |
| 21. (2) | 22. (1) | 23. (1) | | |

Multiple Correct Answers Type

- | | |
|------------------|------------------|
| 1. (1), (2), (3) | 2. (1), (3) |
| 3. (1), (4) | 4. (1), (2), (4) |

5. (1), (4)

6. (1), (2)

7. (1), (2), (3), (4)

8. (2), (4)

9. (2), (4)

Linked Comprehension Type

- | | | | | |
|--------|--------|--------|--------|--------|
| 1. (2) | 2. (1) | 3. (3) | 4. (3) | 5. (1) |
| 6. (4) | | | | |

Numerical Value Type

- | | | | | |
|--------|--------|--------|--------|----------|
| 1. (2) | 2. (3) | 3. (2) | 4. (0) | 5. (1.5) |
|--------|--------|--------|--------|----------|

Chapter 3

Concept Application Exercises

Exercise 3.1

- The given pair of lines is $(2x + y)(x - y) = 0$.
So, the required pair of lines is
 $(2x + y + k)(x - y + k') = 0$
where $2x + y + k = 0$ and $x - y + k' = 0$ pass through $(1, 0)$. Therefore,
 $k = -2, k' = -1$
Therefore, the required pair of lines is
 $(2x + y - 2)(x - y - 1) = 0$
or $2x^2 - y^2 - xy - 4x + y + 2 = 0$
- The given equation is
 $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$
Writing (i) as a quadratic equation in x , we get
 $2x^2 + (5y + 6)x + 3y^2 + 7y + 4 = 0$
or $x = \frac{-(5y + 6) \pm \sqrt{(5y + 6)^2 - 4 \times 2(3y^2 + 7y + 4)}}{4}$
 $= \frac{-(5y + 6) \pm \sqrt{25y^2 + 60y + 36 - 24y^2 - 56y - 32}}{4}$
 $= \frac{-(5y + 6) \pm \sqrt{y^2 + 4y + 4}}{4} = \frac{-(5y + 6) \pm (y + 2)}{4}$
 $\therefore x = \frac{-5y - 6 + y + 2}{4}, \frac{-5y - 6 - y - 2}{4}$
or $4x + 4y + 4 = 0$ and $4x + 6y + 8 = 0$
or $x + y + 1 = 0$ and $2x + 3y + 4 = 0$
Hence, (1) represents a pair of straight lines whose equations are
 $x + y + 1 = 0$ (1)
and $2x + 3y + 4 = 0$ (2)
Solving these two equations, the required point of intersection is $(1, -2)$.
Angle between lines,
 $\tan \theta = \left| \frac{-1 - \left(-\frac{2}{3}\right)}{1 + (-1)\left(-\frac{2}{3}\right)} \right| = \frac{1}{5}$. So, $\theta = \tan^{-1} \frac{1}{5}$
- The bisector of the angle between the positive directions of the axes is $y = x$.
Since it is one of the lines of the given pair of lines $ax^2 + 2hxy + by^2 = 0$, we have
 $x^2(a + 2h + b) = 0$ or $a + b = -2h$.
- The given equation of pair of straight lines can be rewritten as
 $(\sqrt{3}x - y)(x - \sqrt{3}y) = 0$
Their separate equations are
 $y = \sqrt{3}x$ and $y = \frac{1}{\sqrt{3}}x$
or $y = \tan 60^\circ x$ and $y = \tan 30^\circ x$
After rotation, the separate equations are
 $y = \tan 90^\circ x$ and $y = \tan 60^\circ x$
or $x = 0$ and $y = \sqrt{3}x$
Therefore, the combined equation of lines in the new position is
 $x(\sqrt{3}x - y) = 0$ or $\sqrt{3}x^2 - xy = 0$

- The given equation will represent a pair of real and distinct lines if
 $h^2 > ab$
i.e., $\left(-\frac{k}{2}\right)^2 > (2)(2)$
or $k^2 > 16$
or $(k - 4)(k + 4) > 0$
i.e., $k \in (-\infty, -4) \cup (4, \infty)$
- Let $\phi(x, y) = 6x^2 + 5xy - 21y^2 + 13x + 38y - 5 = 0$.
Differentiating with respect to x treating y as constant, we get
 $\frac{d\phi}{dx} = 12x + 5y + 13$
Differentiating with respect to y treating x as constant, we get
 $\frac{d\phi}{dy} = 5x - 42y + 38$
Solving equations $12x + 5y + 13 = 0$ and $5x - 42y + 38 = 0$, we get
 $x = -\frac{32}{23}$ and $y = \frac{17}{23}$
Therefore, the point of intersection is $(-32/23, 17/23)$.

Exercise 3.2

- If m and $2m$ are the slopes, then
 $m + 2m = -\frac{2/h}{1/b} = -\frac{2b}{h}$
and $m \times 2m = \frac{1/a}{1/b} = \frac{b}{a}$
Eliminating m , we get
 $2\left(-\frac{2b}{3h}\right)^2 = \frac{b}{a}$
or $\frac{ab}{h^2} = \frac{9}{8}$
- Let ϕ be the angle between the lines represented by
 $x^2 + 2xy \sec \theta + y^2 = 0$ (1)
Here, $a = 1, b = 1$, and $h = \sec \theta$. Hence,
 $\tan \phi = \frac{2\sqrt{\sec^2 \theta - 1} \times 1}{1 + 1}$
 $= \frac{2\sqrt{\sec^2 \theta - 1}}{2} = \tan \theta$
 $\therefore \phi = \theta$
Hence, the angle between the lines is θ .
- The equation of the line is
 $3x - 2y = 1$ (1)
and the equation of the curve is
 $3x^2 + 5xy - 3y^2 + 2x + 3y = 0$ (2)
Making (2) homogenous with the help of (1), we get
 $3x^2 + 5xy - 3y^2 + (2x + 3y)(3x - 2y) = 0$
or $9x^2 + 10xy - 9y^2 = 0$ (3)
which is the equation of the lines joining the origin to the point of intersection of (1) and (2).
Here, $a = 9, b = -9$.
Since $a + b = 0$, the two lines given by (3) are at right angles.
- Writing the given equation as quadratic in x , we have
 $6x^2 + (5y + 7)x - (4y^2 - 13y + 3) = 0$
or $x = \frac{-(5y + 7) \pm \sqrt{(5y + 7)^2 + 24(4y^2 - 13y + 3)}}{12}$

$$= \frac{-(5y+7) \pm \sqrt{121y^2 - 242y + 121}}{12}$$

$$= \frac{-5(y+7) \pm 11(y-1)}{12} = \frac{6y-18}{12}, \frac{-16y+4}{12}$$

or $2x - y + 3 = 0$ and $3x + 4y - 1 = 0$

which are the two lines represented by the given equation. The point of intersection is $(-1, 1)$ that is obtained by solving these equations.

Also, $\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}$

$$= \frac{2\sqrt{(5/2)^2 - 6(-4)}}{6 - 4} = \frac{\sqrt{121}}{\sqrt{4}} = \frac{11}{2}$$

($\because a = 6, b = -4, h = 5/2$)

So, the equation of the required line is

$$y - 1 = \frac{11}{2}(x + 1)$$

$$\text{or } 11x - 2y + 13 = 0$$

5. Equation of the given lines is $ax^2 + 2hxy + by^2 = 0$.

Equation of the pair of bisectors is

$$h(x^2 - y^2) = (a - b)xy$$

or $hx^2 - (a - b)xy - hy^2 = 0$ (1)

$\therefore A = h, B = -h, 2H = -(a - b)$

Equation of the pair of bisectors of (1) is

$$H(x^2 - y^2) = (A - B)xy$$

or $-\frac{(a - b)}{2}(x^2 - y^2) = 2hxy$

or $-(a - b)(x^2 - y^2) = 4hxy$

or $(a - b)(x^2 - y^2) + 4hxy = 0$

6. The equation of lines joining the origin to the points of intersection of the given line and curve is

$$5x^2 + 11xy - 8y^2 + (8x - 4y)\left(\frac{x - y}{2}\right) + 12\left(\frac{x - y}{2}\right)^2 = 0$$

or $12x^2 - xy - 3y^2 = 0$

The equation of bisectors is

$$\frac{x^2 - y^2}{12 - (-3)} = \frac{xy}{-1/2}$$

or $x^2 + 30xy - y^2 = 0$

7. The equation of the first pair of lines is

$$2x^2 + 6xy + y^2 = 0$$

The equation of the pair of bisectors is

$$3(x^2 - y^2) = (2 - 1)xy$$

or $3(x^2 - y^2) = xy$ (1)

The equation of the second pair of lines is $4x^2 + 18xy + y^2 = 0$.

The equation of the pair of bisectors is $9(x^2 - y^2) = (4 - 1)xy$

$$9(x^2 - y^2) = 3xy$$

or $3(x^2 - y^2) = xy$ (2)

Equations (1) and (2) are the same.

The given pairs have same angular bisectors.

2. (1) $3a + a^2 - 2 = 0$

or $a^2 + 3a - 2 = 0$

or $a = \frac{-3 \pm \sqrt{9 + 8}}{2} = \frac{-3 \pm \sqrt{17}}{2}$

Therefore, there are two values of a .

3. (1) $9x^2 - 24xy + 16y^2 - 12x + 16y - 12 = 0$

or $(3x - 4y + 2)(3x - 4y - 6) = 0$

Hence, the distance between the lines is $\frac{|6 - (-2)|}{5} = \frac{8}{5}$.

4. (1) Acute angle between the lines $x^2 + 4xy + y^2 = 0$ is $\tan^{-1}\{(2\sqrt{4-1})/(1+1)\} = \tan^{-1}\sqrt{3} = \pi/3$. The angle bisectors of $x^2 + 4xy + y^2 = 0$ are given by

$$\frac{x^2 - y^2}{1 - 1} = \frac{xy}{2}$$

or $x^2 - y^2 = 0$ or $x = \pm y$

As $x + y = 0$ is perpendicular to $x - y = 4$, the given triangle is isosceles with vertical angle equal to $\pi/3$. Hence, it is an equilateral triangle.

5. (2) The straight lines represented by $(y - mx)^2 = a^2(1 + m^2)$ are

$$y - mx = \pm a\sqrt{1 + m^2}$$

i.e., $y - mx = a\sqrt{1 + m^2}$ (1)

and $y - mx = -a\sqrt{1 + m^2}$ (2)

Similarly, the straight lines represented by $(y - nx)^2 = a^2(1 + n^2)$ are

$$y - nx = a\sqrt{1 + n^2}$$
 (3)

and $y - nx = -a\sqrt{1 + n^2}$ (4)

Since the lines (1) and (2) are parallel, the distance between them is

$$\left| \frac{a\sqrt{1 + m^2} + a\sqrt{1 + m^2}}{\sqrt{1 + m^2}} \right| = |2a|$$

Similarly, the lines (3) and (4) are parallel lines and the distance between them is $|2a|$. Since the distances between parallel lines are the same, the four lines form a rhombus.

6. (2) Let $y = mx$ be a line common to the given pairs of lines. Then,

$$am^2 + 2m + 1 = 0 \text{ and } m^2 + 2m + a = 0$$

$$\therefore \frac{m^2}{2(1 - a)} = \frac{m}{a^2 - 1} = \frac{1}{2(1 - a)}$$

or $m^2 = 1$ and $m = -\frac{a + 1}{2}$

or $(a + 1)^2 = 4$

or $a = 1$ or -3

But for $a = 1$, the two pairs have both the lines common. So $a = -3$ and the slope m of the line common to both the pairs is 1. Now,

$$\therefore x^2 + 2xy + ay^2 = x^2 + 2xy - 3y^2 = (x - y)(x + 3y)$$

and $ax^2 + 2xy + y^2 = -3x^2 + 2xy + y^2 = -(x - y)(3x + y)$

So, the equation of the required pair of lines is

$$(x + 3y)(3x + y) = 0$$

or $3x^2 + 10xy + 3y^2 = 0$

7. (1) Let the common line be $y = mx$. Then it must satisfy both the equations. Therefore, we have

$$bm^2 + 2hm + a = 0$$
 (1)

$$b'm^2 + 2h'm + a' = 0$$
 (2)

Exercises

Single Correct Answer Type

1. (2) Being a pair of lines, $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$. This gives $m = 4$. Now, find the angle between the lines.

Solving (1) and (2), we get

$$\frac{m^2}{2(ha' - h'a)} = \frac{m}{ab' - a'b} = \frac{1}{2(bh' - b'h)}$$

Eliminating m , we get

$$\left[\frac{ab' - a'b}{2(bh' - b'h)} \right]^2 = \frac{ha' - h'a}{bh' - b'h}$$

$$\text{or } (ab' - a'b)^2 = 4(ha' - h'a)(bh' - b'h)$$

8. (2) The given equation of the pair of straight lines can be rewritten as

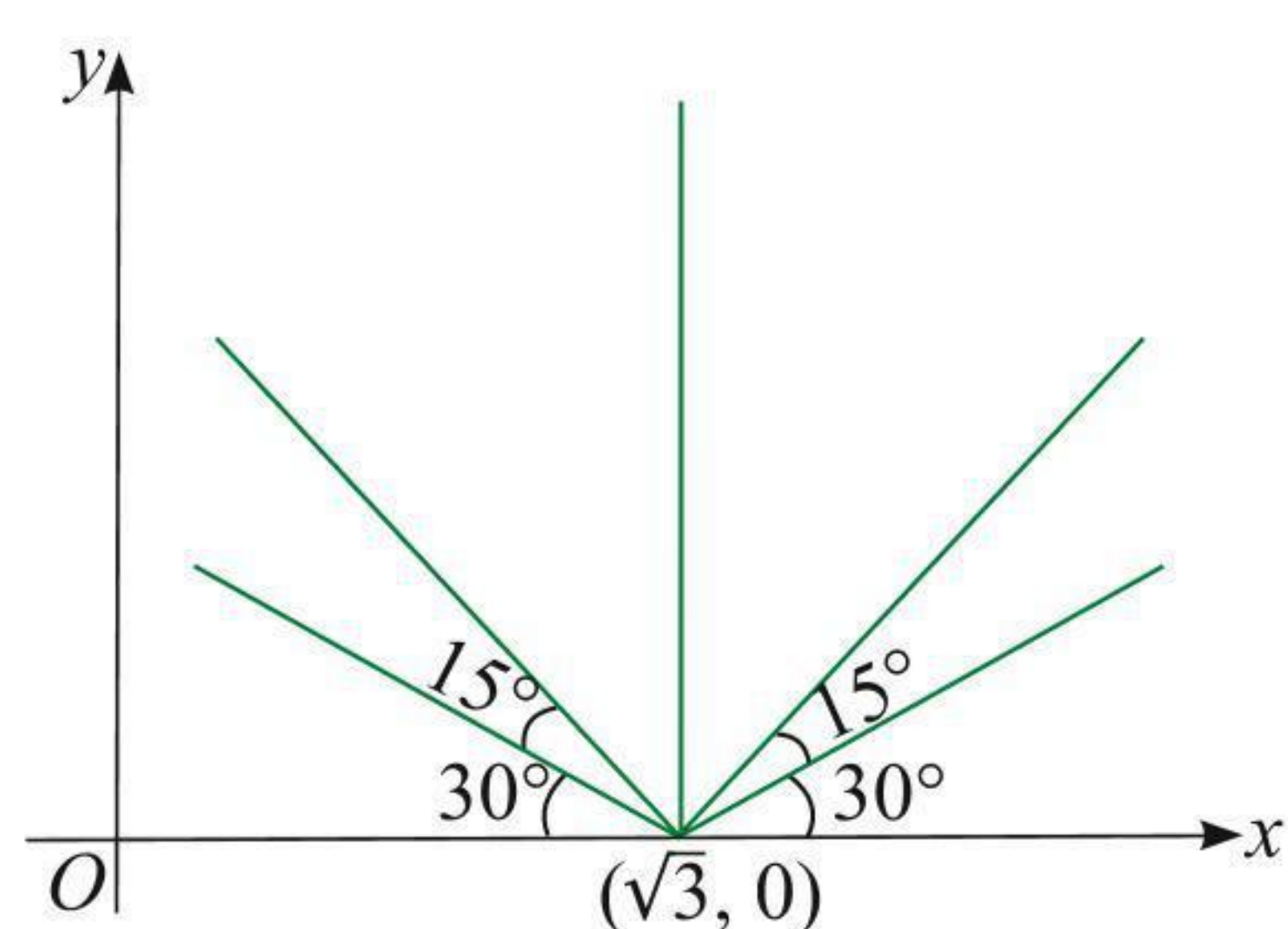
$$(\sqrt{3}y - x + \sqrt{3})(\sqrt{3}y + x - \sqrt{3}) = 0$$

Their separate equations are

$$\sqrt{3}y - x + \sqrt{3} = 0 \text{ and } \sqrt{3}y + x - \sqrt{3} = 0$$

$$\text{or } y = \frac{1}{\sqrt{3}}x - 1 \text{ and } y = -\frac{1}{\sqrt{3}}x + 1$$

$$\text{or } y = (\tan 30^\circ)x - 1 \text{ and } y = (\tan 150^\circ)x + 1$$



After rotation through an angle of 15° as given in the question, the lines are

$$(y - 0) = \tan 45^\circ (x - \sqrt{3}) \text{ and } (y - 0) = \tan 135^\circ (x - \sqrt{3})$$

$$\text{or } y = x - \sqrt{3} \text{ and } y = -x + \sqrt{3}$$

Their combined equation is

$$(y - x + \sqrt{3})(y + x - \sqrt{3}) = 0 \text{ or } y^2 - x^2 + 2\sqrt{3}x - 3 = 0$$

9. (3) We have $6x^2 - xy - 12y^2 = 0$

$$\text{or } (2x - 3y)(3x + 4y) = 0 \quad (1)$$

$$\text{and } 15x^2 + 14xy - 8y^2 = 0$$

$$\text{or } (5x - 2y)(3x + 4y) = 0 \quad (2)$$

Equation of the line common to (1) and (2) is

$$3x + 4y = 0 \quad (3)$$

Equation of any line parallel to (3) is

$$3x + 4y = k$$

Since its distance from (3) is 7, we have

$$\left| \frac{k}{\sqrt{3^2 + 4^2}} \right| = 7 \text{ or } k = \pm 35$$

10. (3) We have

$$x^2y^2 - 9y^2 - 6x^2y + 54y = 0$$

$$\text{or } y^2(x^2 - 9) - 6y(x^2 - 9) = 0$$

$$\text{or } y(y - 6)(x - 3)(x + 3) = 0$$

$$\text{or } y = 0, y = 6, x = 3, x = -3$$

So, the given equation represents four straight lines which form a square.

11. (3) The given equations are

$$a^2x^2 + 2h(a + b)xy + b^2y^2 = 0 \quad (1)$$

$$\text{and } ax^2 + 2hxy + by^2 = 0 \quad (2)$$

The equation of the bisectors of the angles between the lines represented by (1) is

$$\frac{x^2 - y^2}{a^2 - b^2} = \frac{xy}{h(a + b)}$$

$$\text{or } \frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

which is the same as the equation of bisectors of the angles between the line pair (2). Thus, two line pairs are equally inclined to each other.

12. (3) The equations of the given lines are

$$y - 1 = \tan \theta (x - 1)$$

$$\text{and } y - 1 = \cot \theta (x - 1)$$

So, their joint equation is

$$[(y - 1) - \tan \theta (x - 1)][(y - 1) - \cot \theta (x - 1)] = 0$$

$$\text{or } (y - 1)^2 - (\tan \theta + \cot \theta)(x - 1)(y - 1) + (x - 1)^2 = 0$$

$$\text{or } x^2 - (\tan \theta + \cot \theta)xy + y^2 + (\tan \theta + \cot \theta - 2)(x + y - 1) = 0$$

Comparing with the given equation, we get

$$\tan \theta + \cot \theta = a + 2$$

$$\text{or } \frac{1}{\sin \theta \cos \theta} = a + 2$$

$$\text{or } \sin 2\theta = \frac{2}{a + 2}$$

13. (4) Since the product of the slope of the four lines represented by the given equation is 1 and a pair of lines represents the bisectors of the angles between the other two, the product of the slopes of each pair is -1.

So, let the equation of one pair be $ax^2 + 2hxy - ay^2 = 0$. Then the equation of its bisectors is

$$\frac{x^2 - y^2}{2a} = \frac{xy}{h}$$

By hypothesis,

$$x^4 + x^3y + cx^2y^2 - xy^3 + y^4 = (ax^2 + 2hxy - ay^2)(hx^2 - 2axy - hy^2)$$

$$= ah(x^4 + y^4) + 2(h^2 - a^2)(x^3y - xy^3) - 6ahx^2y^2$$

14. (2) Given $AB = BC$. Also,

$$m_1 + m_2 = -\frac{2h}{b} \text{ and } m_1m_2 = \frac{a}{b}$$

$$\tan \theta = \frac{AB}{OA} = m_1$$

$$\text{and } m_2 = \tan \alpha = \frac{2AB}{OA} = 2m_1$$

$$\therefore \frac{m_2}{m_1} = 2$$

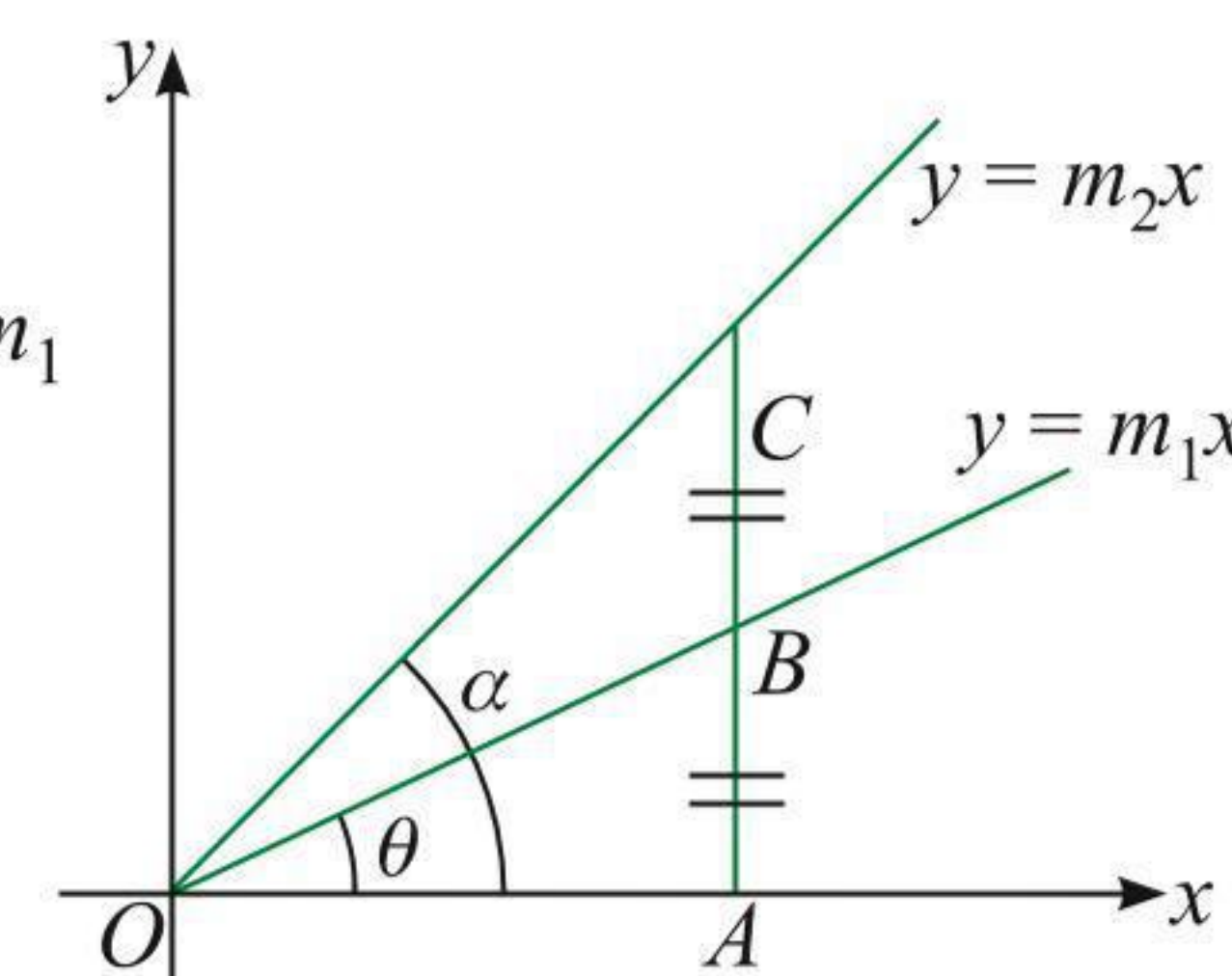
$$\text{or } \frac{m_2 + m_1}{m_2 - m_1} = \frac{2 + 1}{2 - 1} = 3$$

$$\text{or } -\frac{2h/b}{\sqrt{(4h^2/b^2) - (4a/b)}} = 3$$

$$\text{or } \frac{4h^2}{b^2} \cdot \frac{b^2}{4(h^2 - ab)} = 9$$

$$\text{or } 9h^2 - 9ab = h^2$$

$$\text{or } 8h^2 = 9ab$$



15. (4) Each point (x, y) has an image in line $y = 0$ as $(x, -y)$.

So, replacing y by $-y$ in the given equation, we get the image as $ax^2 - 2hxy + by^2 = 0$.

16. (2) The given pair of lines is

$$135x^2 - 136xy + 33y^2 = 0 \quad (1)$$

The equation of bisectors of angles between the pair of lines (1) is

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h} \text{ or } \frac{x^2 - y^2}{135 - 33} = \frac{xy}{-68}$$

$$\text{or } 2x^2 + 3xy - 2y^2 = 0 \quad (2)$$

$$\text{or } (x + 2)(2x - y) = 0$$

One of the bisectors is $x + 2y = 0$ which is parallel to the line $x + 2y = 7$.

Hence, the line $x + 2y = 7$ is equally inclined to the given lines.

17. (2) Let m and m^2 be the slopes of the lines represented by $ax^2 + 2hxy + by^2 = 0$. Then,

$$m + m^2 = -\frac{2h}{b}$$

$$\text{and } mm^2 = \frac{a}{b} \text{ or } m^3 = \frac{a}{b}$$

$$\text{Now, } (m + m^2)^3 = \left(-\frac{2h}{b}\right)^3$$

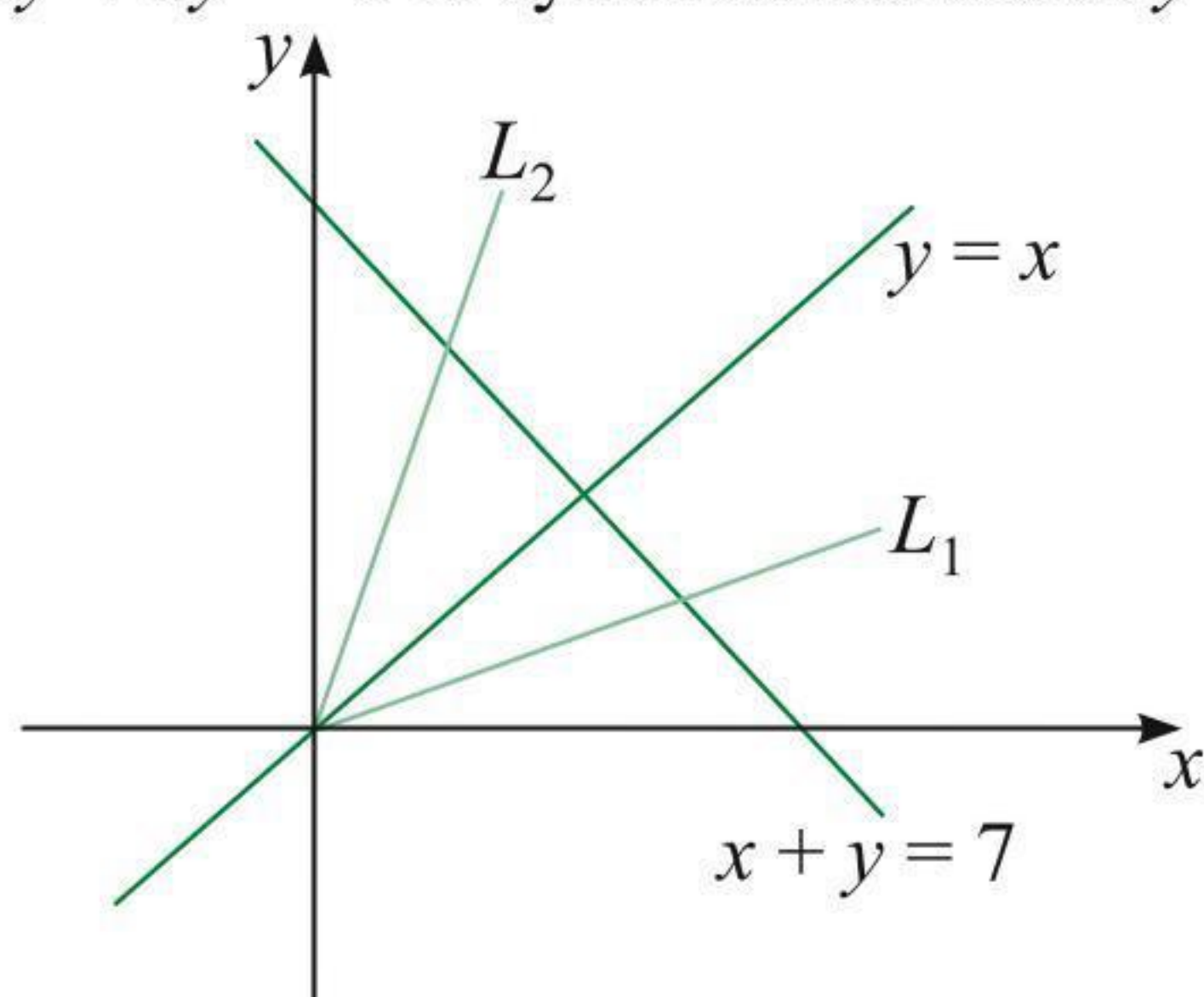
$$\therefore m^3 + m^6 + 3mm^2(m + m^2) = -\frac{8h^3}{b^3}$$

$$\therefore \frac{a}{b^2}(a + b) + \frac{8h^3}{b^3} = \frac{6ah}{b^2}$$

$$\therefore \frac{a + b}{h} + \frac{8h^2}{ab} = 6$$

These are the set of parallel lines and the distance between parallel lines are equal. So, the figure is a rhombus.

18. (1) $ax^2 + 2hxy + ay^2 = 0$ is symmetrical about $y = x$.



19. (2) Put $2h = -(a + b)$ in $ax^2 + 2hxy + by^2 = 0$. Then,

$$ax^2 - (a + b)xy + by^2 = 0$$

$$\text{or } (x - y)(ax - by) = 0$$

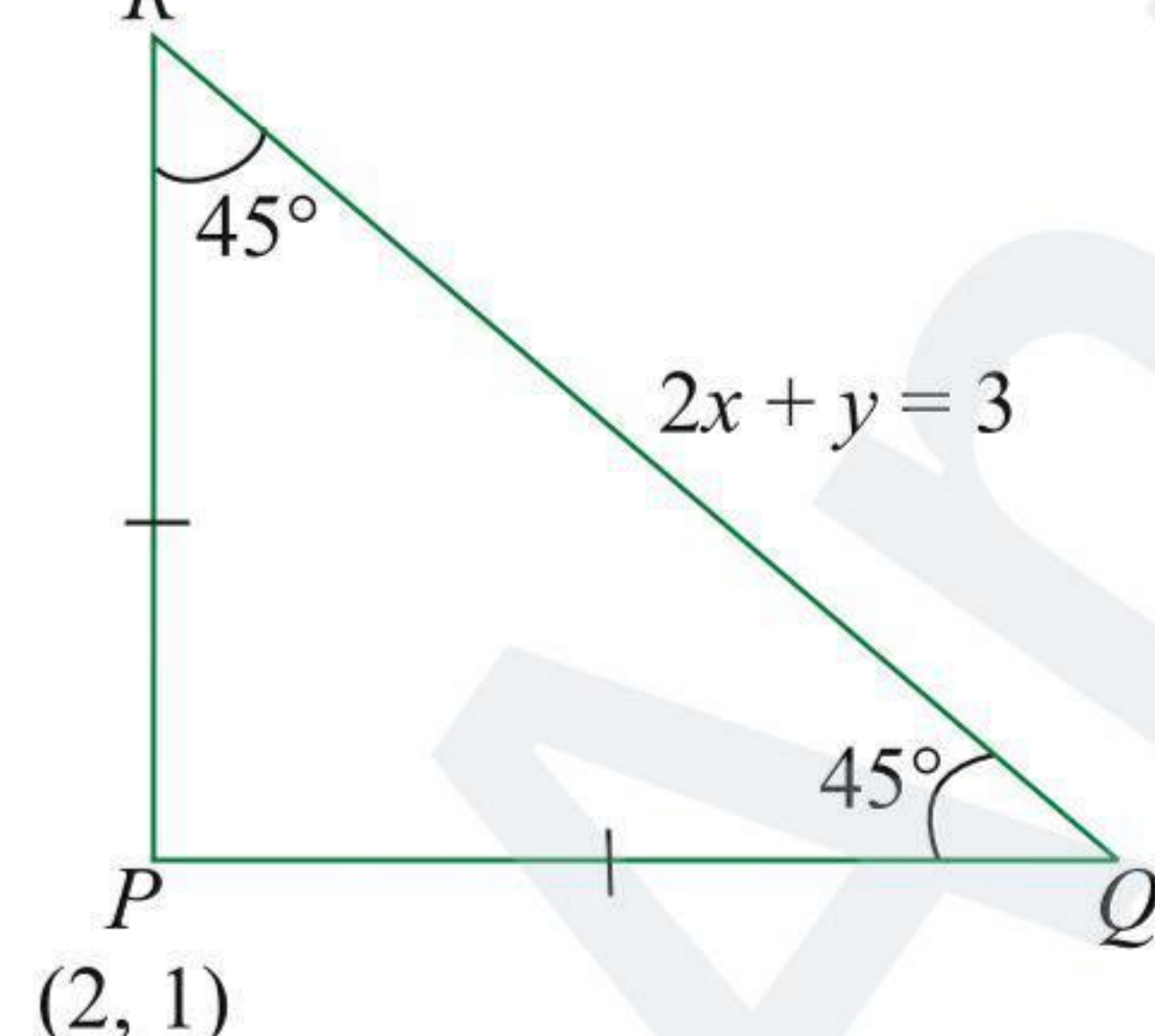
Therefore, one of the lines bisects the angle between the coordinates axes in the positive quadrant. Also, putting $b = -2h - a$ in $ax - by$, we have

$$ax - by = ax - (-2h - a)y = ax + (2h + a)y$$

Hence, $ax + (2h + a)y$ is a factor of $ax^2 + 2hxy + by^2$. However, statement 2 is not the correct explanation of statement 1.

20. (3) The lines by which triangle is formed are $x = 0$, $y = 0$, and $x + y = 1$. Clearly, it is a right triangle. We know that in a right-angled triangle, the orthocenter coincides with the vertex at which right angle is formed. Therefore, the orthocenter is $(0, 0)$.

21. (2)



Let m be the slope of PQ . Then,

$$\tan 45^\circ = \left| \frac{m - (-2)}{1 + m(-2)} \right| = \left| \frac{m + 2}{1 - 2m} \right|$$

$$\therefore m + 2 = 1 - 2m \text{ or } -1 + 2m = m + 2$$

$$\therefore m = -\frac{1}{3} \text{ or } m = 3$$

Hence, the equation of PQ is

$$y - 1 = -\frac{1}{3}(x - 2)$$

$$\text{or } x + 3y - 5 = 0$$

and the equation of PR is

$$y - 1 = 3(x - 2)$$

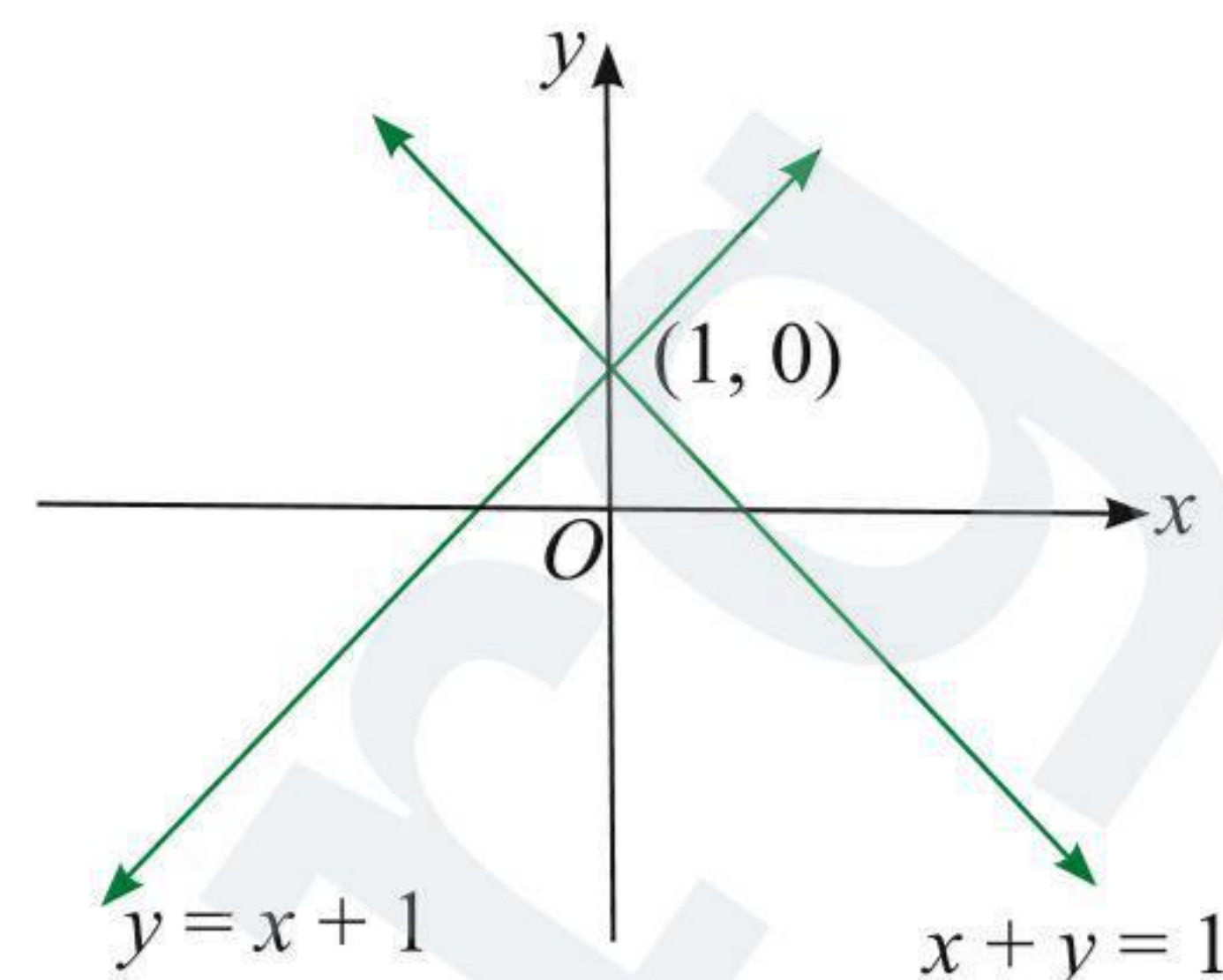
$$\text{or } 3x - y - 5 = 0$$

Hence, the combined equation of PQ and PR is

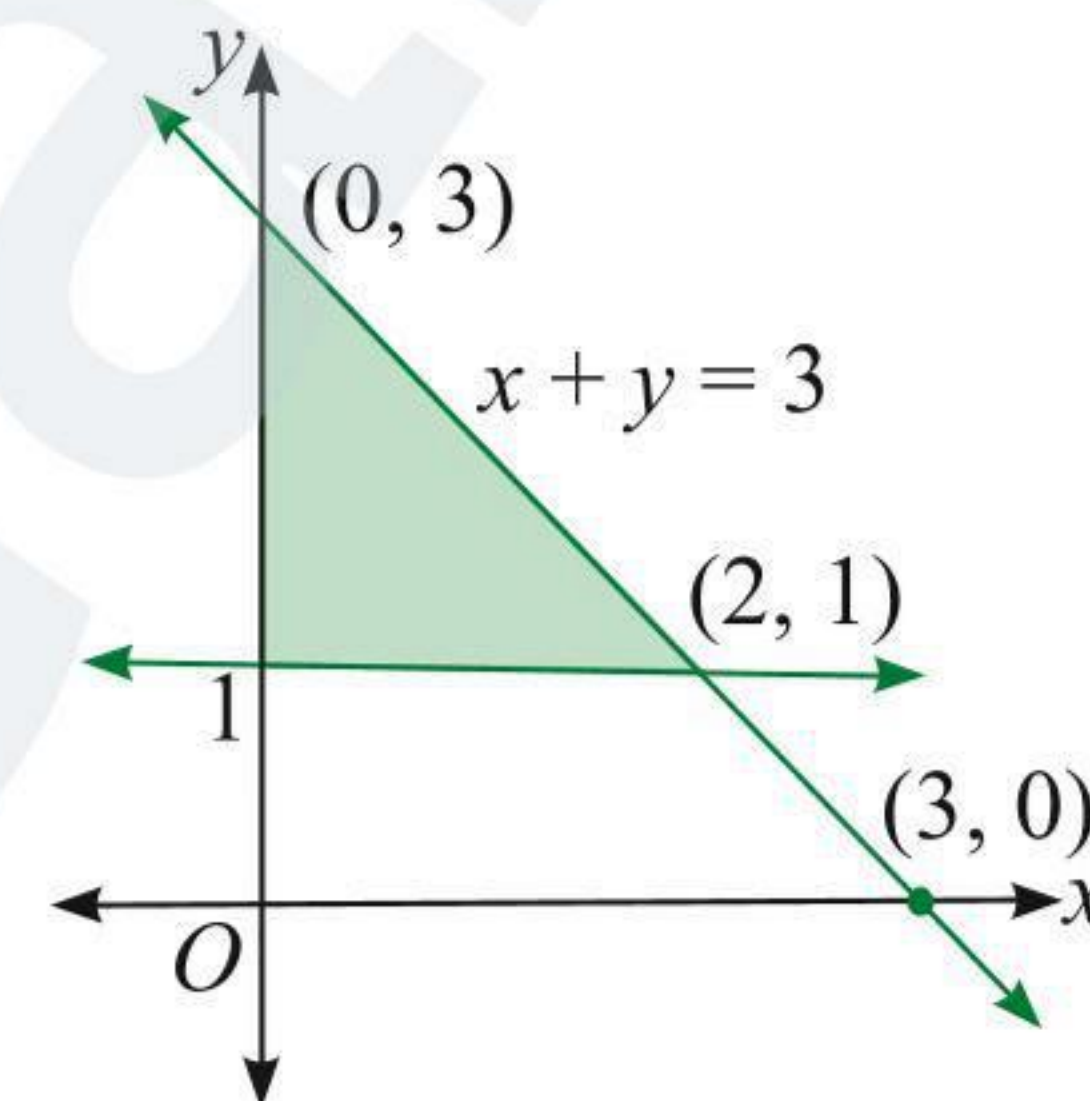
$$(x + 3y - 5)(3x - y - 5) = 0$$

$$\text{or } 3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$$

22. (1) $x^2 - y^2 + 2y = 1$ or $x = \pm(y - 1)$



The bisectors of the above lines are $x = 0$ and $y = 1$.



So, the area between $x = 0$, $y = 1$, and $x + y = 3$ is the shaded region shown in the figure. The area is given by $(1/2) \times 2 \times 2 = 2$ sq. units.

23. (1) The component lines of given pair of straight lines are perpendicular to each other. So, orthocentre is the point of intersection of component lines.

Multiple Correct Answers Type

1. (1), (2), (3)

The equation is

$$x^2(x + y) - y^2(x + y) = 0$$

$$\text{or } (x + y)^2(x - y) = 0$$

It represents the lines $x + y = 0$, $x + y = 0$, $x - y = 0$.

2. (1), (3)

The equation represents a pair of straight lines. Hence,

$$1 \times (-2)(-1) + 2\left(\frac{3}{2}\right) \times 0 \times \frac{m}{2} - 1 \times \left(\frac{3}{2}\right)^2 - (-2)$$

$$\times 0^2 - (-1) \times \left(\frac{m}{2}\right)^2 = 0$$

$$\text{or } m = 1, -1$$

The points of intersection of the pair of lines are obtained by solving

$$\frac{\partial S}{\partial x} \equiv 2x + my = 0$$

$$\text{and } \frac{\partial S}{\partial y} \equiv mx - 4y + 3 = 0$$

When $m = 1$, the required point is the intersection of $2x + y = 0$, $x - 4y + 3 = 0$. When $m = -1$, the required point is the intersection of $2x - y = 0$, $-x - 4y + 3 = 0$.

3. (1), (4)

$$\text{Here, } my(y - mx) + x(y - mx) = 0$$

$$\text{or } (y - mx)(my + x) = 0$$

So, the lines are $y = mx$ or $y = (-1/m)x$. The bisectors between the lines $xy = 0$ are $y = x$ and $y = -x$. Therefore, $m = 1, -1$.

4. (1), (2), (4)

Let the equation of the lines given by $x^2 + 2hxy + y^2 = 0$ be $y = m_1x$ and $y = m_2x$. Since these make an angle α with $y + x = 0$ whose slope is -1 , we have

$$\frac{m_1 + 1}{1 - m_1} = \tan \alpha = \frac{-1 - m_2}{1 - m_2}$$

$$\text{or } m_1 + m_2 = \frac{(\tan \alpha - 1)^2 + (\tan \alpha + 1)^2}{\tan^2 \alpha - 1}$$

$$= \frac{-2 \sec^2 \alpha \times \cos^2 \alpha}{\cos 2\alpha}$$

$$\therefore -2 \sec 2\alpha = -2h$$

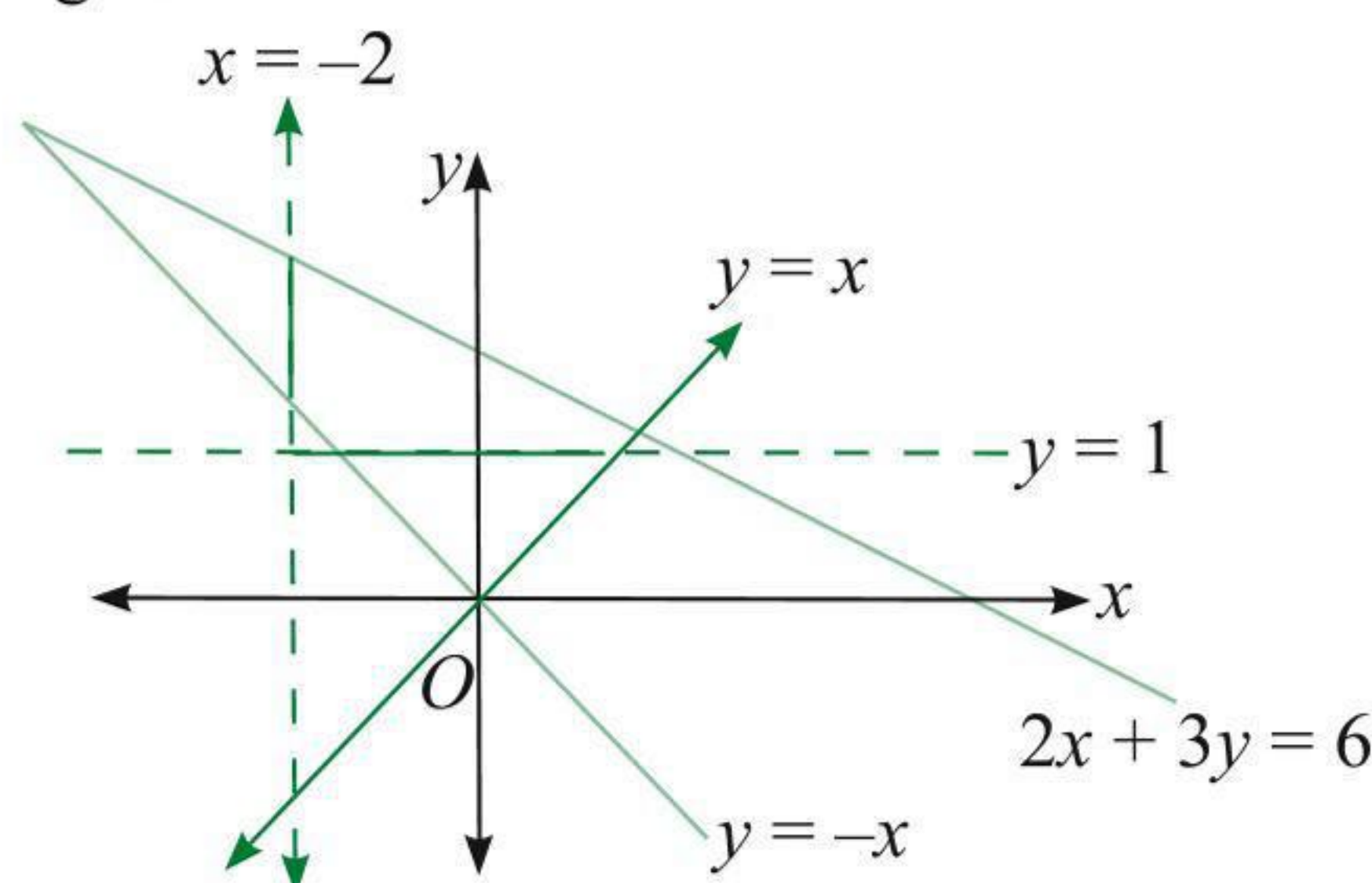
$$\text{or } \sec 2\alpha = h$$

$$\text{or } \cos 2\alpha = \frac{1}{h} \text{ or } 2 \cos^2 \alpha - 1 = \frac{1}{h}$$

$$\text{or } \cos \alpha = \sqrt{\frac{1+h}{2h}} \text{ and } \cot \alpha = \sqrt{\frac{h+1}{h-1}}$$

5. (1), (4)

The separate equations of the sides are $x - y = 0$, $x + y = 0$, and $2x + 3y - 6 = 0$. The point $(-2, a)$ moves on the line $x = -2$ and the point $(b, 1)$ moves on the line $y = 1$. From the figure, the y -coordinates of the points of intersection of $x = -2$ with $y = -x$ and $2x + 3y = 6$ give the range of values of a . The x -coordinates of the points of intersection of $y = 1$ with $y = -x$ and $y = x$ give the range of values of b .



6. (1), (2)

$$\text{Let } \frac{2}{3}x^2 + \frac{p}{3}xy + y^2 = (y - mx)(y - m'x)$$

$$\text{and } \frac{2}{-3}x^2 + \frac{q}{-3}xy + y^2 = (y + \frac{1}{m}x)(y - m'x)$$

Then,

$$m + m' = -\frac{p}{3}, mm' = \frac{2}{3} \quad (1)$$

$$\frac{1}{m} - m' = \frac{-q}{3}, -\frac{m'}{m} = -\frac{2}{3} \quad (2)$$

$$\text{or } m^2 = 1 \text{ or } m = \pm 1$$

If $m = 1$, $m' = 2/3$ and, so, $p = -5$, $q = -1$. If $m = -1$, $m' = -2/3$ and, so, $p = 5$, $q = 1$.

7. (1), (2), (3), (4)

The equation of the lines joining the origin to the points of intersection of the given lines is

$$3x^2 + mxy - 4x(2x + y) + 1(2x + y)^2 = 0 \quad (\text{by homogenization})$$

$$\text{or } x^2 - mxy - y^2 = 0$$

which are perpendiculars for all values of m .

8. (2), (4)

m and m^2 are the roots of equation

$$\left(\frac{y}{x}\right)^2 - 6\left(\frac{y}{x}\right) + a = 0$$

$$\therefore m + m^2 = 6 \text{ and } m^3 = a$$

$$\text{or } m^3 + m^6 + 3m^3(m + m^2) = 216$$

$$a + a^2 + 3a \cdot 6 = 216$$

$$a^2 + 19a - 216 = 0 \text{ or } a = 8, a = -27$$

9. (2), (4)

The equation $y^2 + xy - 12x^2 = 0$ can be rewritten as

$$(y + 4x)(y - 3x) = 0$$

$$\text{or } \frac{y}{x} = -4, 3$$

The two pairs will have a line common if -4 or 3 is a root of

$$b\left(\frac{y}{x}\right)^2 + 2h\left(\frac{y}{x}\right) + a = 0$$

$$\therefore 9b + 6h + a = 0 \text{ or } 16b - 8h + a = 0$$

Linked Comprehension Type

1. (2) $a = \lambda$; $h = -5$; $b = 12$; $g = \frac{5}{2}$; $f = -8$, $c = -3$

$$\therefore \lambda(12)(-3) + 2(-8)\left(\frac{5}{2}\right)(-5) - \lambda(64) - \left(\frac{25}{4}\right) \times 12 + 3 \times 25 = 0$$

$$\text{or } -36\lambda + 200 - 64\lambda - 75 + 75 = 0$$

$$\text{or } 100\lambda = 200$$

$$\text{or } \lambda = 2$$

2. (1) $2x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$

Consider the homogeneous part

$$2x^2 - 10xy + 12y^2 = (x - 2y)(2x - 6y)$$

$$\therefore 2x^2 - 10xy + 12y^2 + 5x - 16y - 3$$

$$\equiv (2x - 6y + A)(x - 2y + B)$$

Comparing coefficients, we get

$$A = -1, B = 3$$

Hence, the lines are

$$2x - 6y - 1 = 0 \quad \text{and} \quad x - 2y + 3 = 0$$

Solving, we get the intersection point as $(-10, -7/2)$. Therefore,

Product = 35

3. (3) $\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b} = \frac{2\sqrt{25 - 24}}{14} = \frac{1}{7}$

4. (3), 5. (1), 6. (4)

As the lines are perpendicular,

Coefficient of x^2 + Coefficient of $y^2 = 0$

$$\therefore a - 2 = 0$$

$$\text{or } a = 2$$

$$\text{Also, } abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

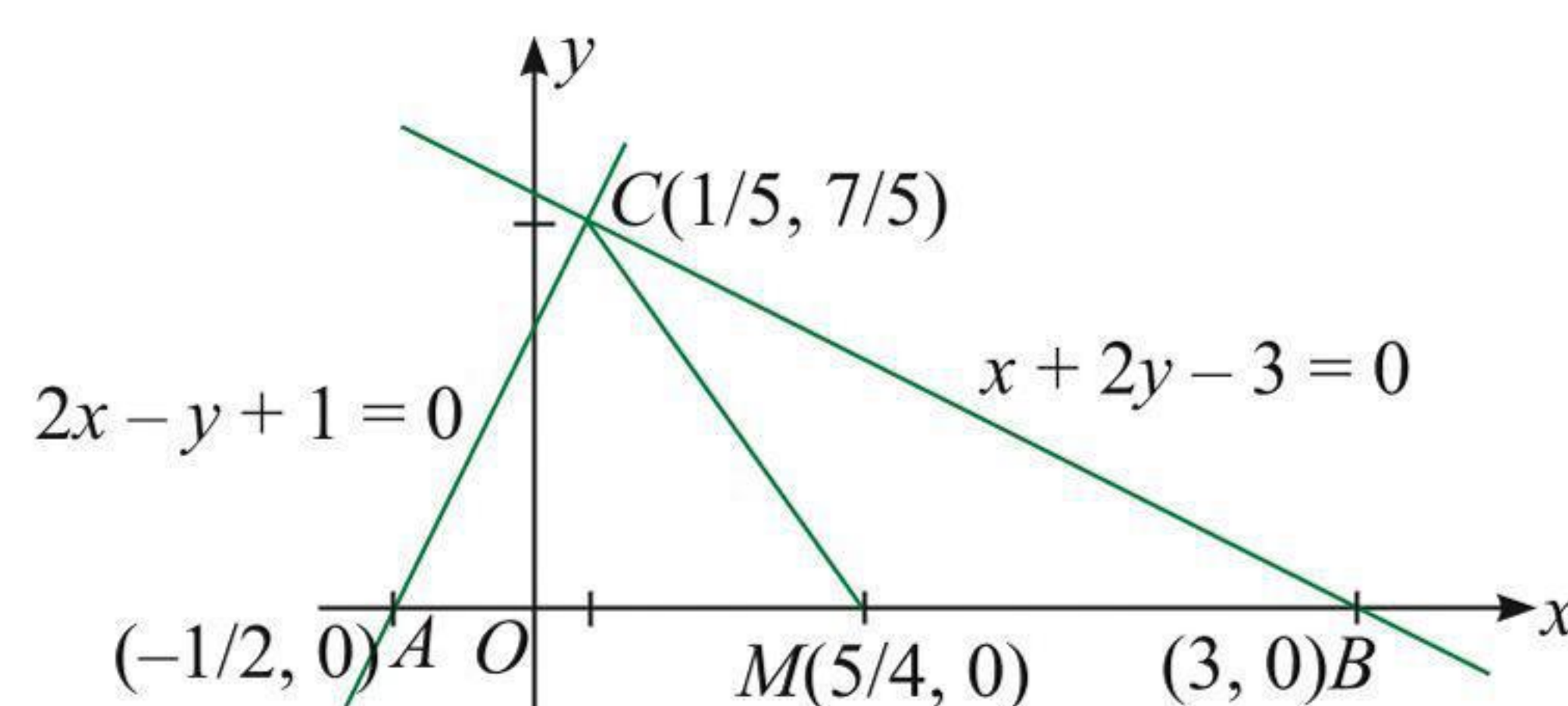
$$\therefore c = -3$$

Hence, the given pair of lines is

$$2x^2 + 3xy - 2y^2 - 5x + 5y - 3 = 0$$

Factorizing, we get lines

$$x + 2y - 3 = 0 \text{ and } 2x - y + 1 = 0$$



The point of intersection of the lines is $C(1/5, 7/5)$.

The points of intersection of the lines with the x -axis are $A(3, 0)$ and $B(-1/2, 0)$.

The orthocenter of triangle is $C(1/5, 7/5)$ and the circumcenter is the midpoint of AB which is $M(5/4, 0)$. Therefore,

$$CM = \sqrt{\left(\frac{5}{4} - \frac{1}{5}\right)^2 + \frac{49}{25}} = \frac{7}{4}$$

Numerical Value Type

1. (2) $(x + 7y)^2 + 4\sqrt{2}(x + 7y) - 42 = 0$

or $(x + 7y)^2 + 7\sqrt{2}(x + 7y) - 3\sqrt{2}(x + 7y) - 42 = 0$

or $(x + 7y)[x + 7y + 7\sqrt{2}] - 3\sqrt{2}(x + 7y + 7\sqrt{2}) = 0$

or $(x + 7y + 7\sqrt{2})(x + 7y - 3\sqrt{2}) = 0$

or $x + 7y + 7\sqrt{2} = 0$ and $x + 7y - 3\sqrt{2} = 0$

or $d = \left| \frac{7\sqrt{2} + 3\sqrt{2}}{\sqrt{1 + 49}} \right| = \frac{10\sqrt{2}}{\sqrt{50}} = 2$

2. (3) We have

$$y^2 - 9xy + 18x^2 = 0$$

or $y^2 - 6xy - 3xy + 18x^2 = 0$

or $y(y - 6x) - 3x(y - 6x) = 0$

or $y - 3x = 0$ and $y - 6x = 0$

The third line is $y = 6$. Therefore, the area of the triangle formed by these lines

$$= \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 1 & 6 & 1 \\ 2 & 6 & 1 \end{vmatrix}$$

$$= \frac{1}{2} |6 - 12|$$

$$= 3 \text{ units}^2$$

3. (2) The equation $4x^2 + 8xy + ky^2 - 9 = 0$ represents a pair of straight lines if $(4)(k)(-9) - (-9)(4)^2 = 0$ or $k = 4$.

4. (0) Let m_1 and m_2 be the slopes of the lines $x^2 + hxy + 2y^2 = 0$. Then,

$$m_1 + m_2 = -\frac{h}{2}, m_1 m_2 = \frac{1}{2}$$

But $m_1 = 2m_2$ (given). Therefore,

$$3m_2 = -\frac{h}{2} \text{ and } 2m_2^2 = \frac{1}{2}$$

i.e., $m_2^2 = \frac{1}{4}$

Also, $m_2 = -\frac{h}{6}$

$$\therefore \frac{h^2}{36} = \frac{1}{4} \text{ or } h^2 = 9 \text{ or } h = \pm 3$$

5. (1.5)

Shifting the origin to $(1, 2)$, the equation of given pair of lines becomes

$$ax^2 + 2hxy + by^2 = 0$$

whose one angle bisector is $x + 2y = 0$ (given) and other is $2x - y = 0$.

$\therefore$ Combined equation of angle bisector is

$$(x + 2y)(2x - y) = 0$$

On comparing with $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$, we get

$$a - b = -3 \text{ and } h = 2$$

$$\therefore \frac{b - a}{h} = \frac{3}{2}$$